

Chapter 1

Python + Biology: Your First Programs

Start with print, variables, simple data types, input, and tiny DNA examples.

EDOS learning idea

We will use biology and DNA examples because they are concrete, visual, and easy to experiment with. Type the examples yourself, change one thing, run again, and observe what changed.

What you will be able to do

- Understand what Python is and why biologists use it.
- Run simple Python statements and read the output.
- Store information in variables.
- Recognize strings, integers, floats, and booleans.
- Use small DNA examples without feeling overwhelmed.

Classroom rhythm

Read a small idea -> run the example -> change one value -> predict the result -> run again -> explain what happened in your own words.

1.1 What is Python?

Python is a programming language. In plain language, it is a way to write instructions that a computer can follow. The nice part is that Python reads more like normal English than many older programming languages.

In biology, Python can help you count DNA bases, organize sample names, clean spreadsheet-like data, draw graphs, compare sequences, and automate repetitive work. You do not need to become a software engineer before Python becomes useful.

A good mental model: imagine Python as a very fast lab assistant. It follows your instructions exactly. If the instruction is clear, it works quickly. If the instruction is unclear, it asks you to fix it with an error message.

1.2 Your first line: `print()`

The `print()` function sends information to the output area. It is one of the easiest ways to check what your program is doing.

```
print('Hello, biology!')
print('DNA is information stored in molecules.')
```

Text inside quotation marks is called a string. Python keeps the characters exactly as text.

1.3 Variables are labeled sample tubes

A variable is a name that points to a value. Think of a labeled tube: the label tells you what is inside. In code, the variable name is the label.

```
organism = 'E. coli'
colonies = 24
temperature = 37.0
is_control = True

print(organism)
print(colonies)
```

Here `organism` is a string, `colonies` is an integer, `temperature` is a float, and `is_control` is a boolean. Boolean values are simply `True` or `False`.

1.4 Python can do quick calculations

Python can act like a calculator, but variables make calculations easier to understand.

```
a_count = 20
t_count = 21
total_at = a_count + t_count
print(total_at)
print(a_count / total_at)
```

Useful operators include +, -, *, /, //, %, and **. You can experiment safely in EDOS.

1.5 Optional keyboard input

The `input()` function asks the user to type something. Remember: `input()` gives you text unless you convert it.

```
name = input('What organism are you studying? ')
print('Today we are studying', name)
```

For classroom exercises, fixed values are often simpler at first. Input becomes useful when you want interactive programs.

1.6 First DNA mini-program

DNA sequences are perfect beginner examples because they are just strings made from A, C, G, and T.

```
dna = 'AGCTTTTCATTCTGAC'
print('Sequence:', dna)
print('Number of bases:', len(dna))
```

`len()` means length. It works on strings, lists, and many other Python objects. We will use it constantly.

Try it yourself

These are practice prompts, not a test. If your code breaks, that is useful information. Read the error, change one thing, and try again.

1. Change the organism variable to your favorite organism.
2. Create variables for A, C, G, and T counts, then add them together.
3. Store a short DNA sequence in dna and print its length with len().
4. Make one intentional typo, run the code, and read the error instead of being afraid of it.

Tiny debugging habit

When something fails, check spelling, quotation marks, brackets, colons, and indentation first. Then print intermediate values so you can see what Python sees.

Chapter summary

- Python is a language for giving computers instructions.
- print() shows output.
- Variables give useful names to values.
- Strings store text such as DNA sequences.
- len() tells you how many items or characters are inside something.

Next step

Open Notebook 01 in EDOS Jupyter and run the matching exercises. Then continue to Chapter 2.

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